System Analysis

DR. DRAGUNOV

Project Managment

Q1 V1

Depict graphically: **Phased development methodology.**

During Project Planning phase, next cash flows were identified:

Development costs

- Office space and equipment: $40 000
- Vendor installation: $50 000
- Data conversion tools: $10 000

Operational costs

- Software upgrades: $20 000
- Operational team salaries: $ 10 000
- User training: $ 20 000

Tangible benefits

- Increased sales: $20 000
- Reduction in staff: $40 000
- Better supplier prices: $30 000

Find: ROI, BEP and NPV, if project life cycle is 3 years and rate of return = 5%

Q1 V2

Depict graphically: **Waterfall development methodology.**

During Project Planning phase, next cash flows were identified:

Development costs

- Office space and equipment: $40 000
- Vendor installation: $50 000
- Data conversion tools: $20 000

Operational costs

- Software upgrades: $20 000
- Operational team salaries: $ 10 000
- User training: $ 10 000

Tangible benefits

- Increased sales: $40 000
- Reduction in staff: $40 000
- Better supplier prices: $30 000

Find: ROI, BEP and NPV, if project life cycle is 3 years and rate of return = 10%

Q1 V3

Depict graphically: **Parallel development methodology.**

During Project Planning phase, next cash flows were identified:

Development costs

- Office space and equipment: $40 000
- Vendor installation: $50 000
- Data conversion tools: $30 000

Operational costs

- Software upgrades: $20 000
- Operational team salaries: $ 10 000
- User training: $ 10 000

Tangible benefits

- Increased sales: $60 000
- Reduction in staff: $40 000
- Better supplier prices: $30 000

Find: ROI, BEP and NPV, if project life cycle is 3 years and rate of return = 15%

Q1 V4

Depict graphically: **Phased development methodology.**

During Project Planning phase, next cash flows were identified:

Development costs

- Office space and equipment: $40 000
- Vendor installation: $50 000
- Data conversion tools: $40 000

Operational costs

- Software upgrades: $20 000
- Operational team salaries: $ 10 000
- User training: $ 10 000

Tangible benefits

- Increased sales: $80 000
- Reduction in staff: $40 000
- Better supplier prices: $30 000

Find: ROI, BEP and NPV, if project life cycle is 3 years and rate of return = 20%

PROJECT SELECTION AND MANAGEMENT



Introduction

Most IT departments face a demand for IT projects that far exceeds the department's ability to supply them. In the past 10 years, business application growth has exploded, and chief information officers (CIOs) are challenged to select projects that will provide the highest possible return on IT investments while managing project risk. In a recent analysis, AMR Research, Inc., found that 2–15% of projects taken on by IT departments are not strategic to the business. In today's globally competitive business environment, the corporate IT department needs to carefully prioritize, select, and manage its portfolio of development projects.

Once selected, a systems development project undergoes a thorough process of project management , the process of planning and controlling the project within a specifi ed time frame, at minimum cost, with the desired outcomes. A project manager has the primary responsibility for managing the hundreds of tasks and roles that need to be carefully coordinated. Project management has evolved into an actual profession with many training options and professional certifi cation (e.g., project management professional [PMP]) available through the Project Management Institute (www.pmi.org).

Ways to Classify Projects

Size	What is the size? How many people are needed to work on the project?
Cost	How much will the project cost the organization?
Purpose	What is the purpose of the project? Is it meant to improve the technical infrastructure? Support a current business strategy? Improve operations? Demonstrate a new innovation?
Length	How long will the project take before completion? How much time will go by before value is delivered to the business?
Risk	How likely is it that the project will succeed or fail?
Scope	How much of the organization is affected by the system? A department? A division? The entire corporation?
Return on investment	How much money does the organization expect to receive in return for the amount the project costs?

PROJECT SELECTION PROCESS

4-Step Project Selection Process



PROJECT MANAGEMENT PROCESS STAGES

Stage 1



Project Initiation

- Define the project's **purpose, scope, and objectives**
- Identify **key stakeholders**
- Create **project charter**

Stage 2



Project Planning

- Create a **project plan**
- Create a **work breakdown structure**
- Develop a **risk management strategy**

Stage 3



Project Execution

- **Start working** on the project plan
- Build **workflows, assign tasks** to team members
- Keep everyone **in the loop**

Stage 4



Project Monitoring & Controlling

- Do regular **reviews**
- **Monitor progress** against KPIs
- **Apply changes** when needed

Stage 5



Project Closure

- Complete all **project deliverables**
- Discuss **failures and successes**
- Document **lessons learned**

businessmap

TRADITIONAL PROJECT MANAGEMENT TOOLS

Before we get to actually creating a workplan that is suitable to manage and control **an object-oriented systems development project**, we need to introduce a **set of project management tools** that have been used to successfully manage traditional software development projects (and many other types of projects):

- a work-breakdown structure
- a Gantt chart, and
- a network diagram.

Project Management Tools

Aids in creating workplans

Identify all tasks, their sequence and estimate the time to complete each one

Work breakdown structures (WBS): a hierarchy of tasks to identify:

- Duration of each task
- Current status of each task
- Task dependencies (shows which tasks must be completed before others can begin)

Gantt charts: horizontal bar chart that shows the WBS graphically

Network diagrams: PERT and CPM

A TASK

A **task is a unit of work** that will be performed by a member or members of the development team, such as feasibility analysis. Each task is described by information such as its name, start and completion dates, person assigned to complete the task, deliverables, completion status, priority, resources needed, estimated time to complete the task, and the actual time it took to complete the task

Workplan Information	Example
Name of the task	Perform economic feasibility
Start date	Jan 05, 2010
Completion date	Jan 19, 2010
Person assigned to the task	Project sponsor: Mary Smith
Deliverable(s)	Cost–benefit analysis
Completion status	Open
Priority	High
Resources that are needed	Spreadsheet software
Estimated time	16 hours
Actual time	14.5 hours

Workplan Creation

The first thing a project manager must do is to identify the tasks that need to be accomplished and determine how long each task will take.

Once the tasks have been identified and documented, they are organized within **a work breakdown structure** that is used to drive the creation of **Gantt charts** and **network diagrams** that can be used to graphically portray a traditional workplan. These techniques help a project manager understand and manage the project's progress over time.

Work Breakdown Structures

If a project manager prefers to begin from scratch, he or she can use a structured, topdown approach whereby high-level tasks are first defined and then broken down into subtasks. For example, shows a list of high-level tasks needed to implement a new IT training class.

Task Number	Task Name	Duration (in weeks)	Dependency	Status
1	Identify vendors	2		Complete
2	Review training materials	6	1	Complete
3	Compare vendors	2	2	In Progress
4	Negotiate with vendors	3	3	Open
5	Develop communications information	4	1	In Progress
6	Disseminate information	2	5	Open
7	Create and administer survey	4	6	Open

7.1	Create initial survey	1		Open
7.2	Review initial survey	1	7.1	Open
7.2.1	Review by Director of IT Training	1		Open
7.2.2	Review by Project Sponsor	1		Open
7.2.3	Review by Representative Trainee	1		Open
7.3	Pilot test initial survey	1	7.1	Open
7.4	Incorporate survey changes	1	7.2, 7.3	Open
7.5	Create distribution list	0.5		Open
7.6	Send survey to distribution list	0.5	7.4, 7.5	Open
7.7	Send follow-up message	0.5	7.6	Open
7.8	Collect completed surveys	1	7.6	Open
8	Analyze results and choose vendor	2	4, 7	Open
9	Build new classrooms	11	1	In Progress
10	Develop course options	3	8, 9	Open

A list of tasks hierarchically numbered in this way is called a **work breakdown structure (WBS).** The number of tasks and level of detail depend on the complexity and size of the project. At a minimum, the WBS must include the

- duration of the task,
- the current statuses of the tasks (i.e., open, complete), and
- the task dependencies, which occur when one task cannot be performed until another task is completed.

Gantt Chart

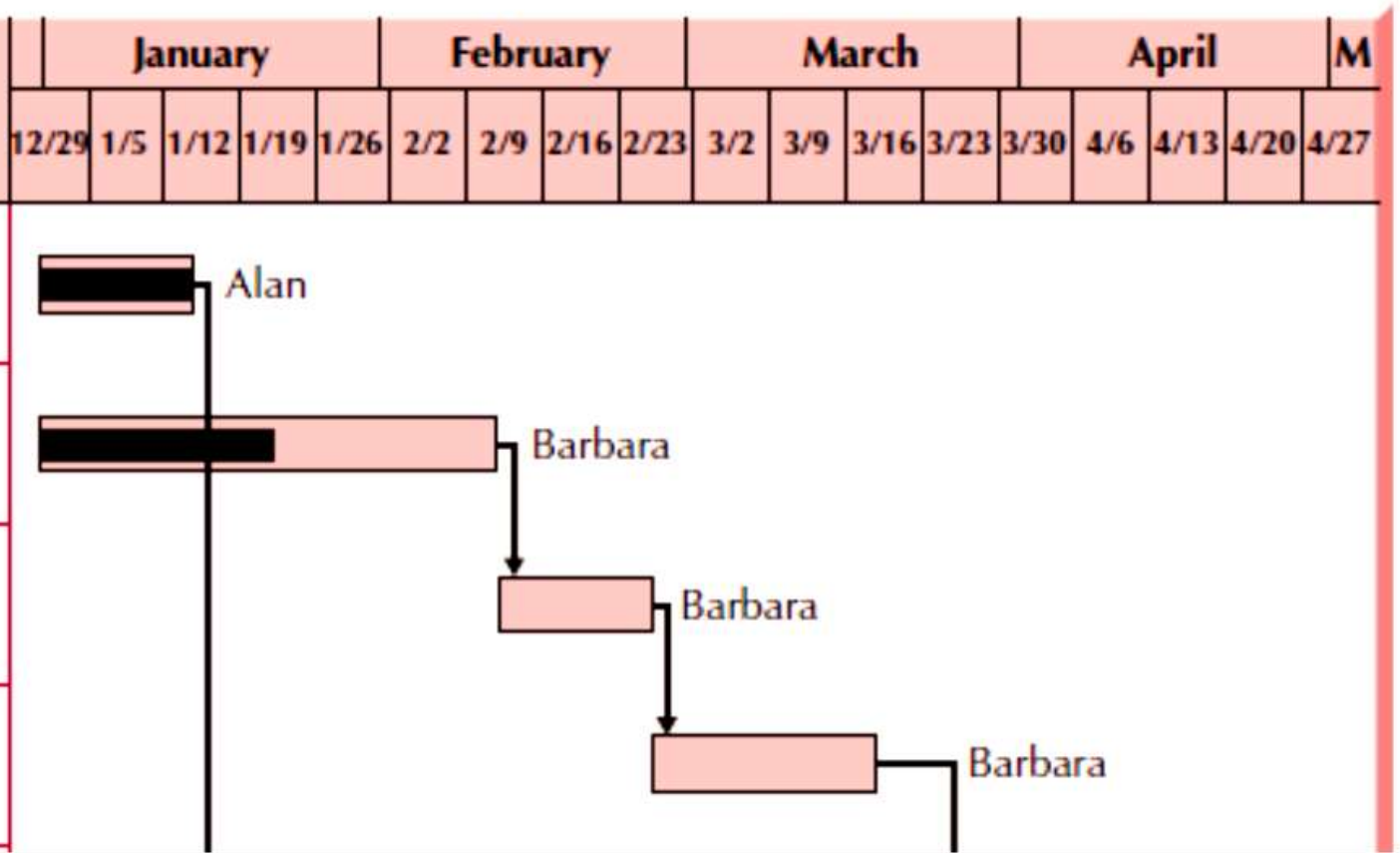
A *Gantt chart* is a horizontal bar chart that shows the same task information as the project WBS but in a graphical way. Sometimes a picture really is worth a thousand words, and the Gantt chart can communicate the high-level status of a project much faster and easier than the WBS. Creating a Gantt chart is simple and can be done using a spreadsheet package, graphics software (e.g., **Microsoft Visio** or **SmartDraw**), or a project management package.



First, tasks are listed as rows in the chart, and time is listed across the top in increments based on the needs of the projects. A short project may be divided into hours or days, whereas a medium-sized project may be represented using weeks or months. Horizontal bars are drawn to represent the duration of each task; the bar's beginning and end mark exactly when the task will begin and end. As people work on tasks, the appropriate bars are filled in proportionately to how much of the task is finished. Too many tasks on a Gantt chart can become confusing, so it's best to limit the number of tasks to around twenty or thirty. If there are more tasks, break them down into subtasks and create Gantt charts for each level of detail.

Grantt Chart example (part)

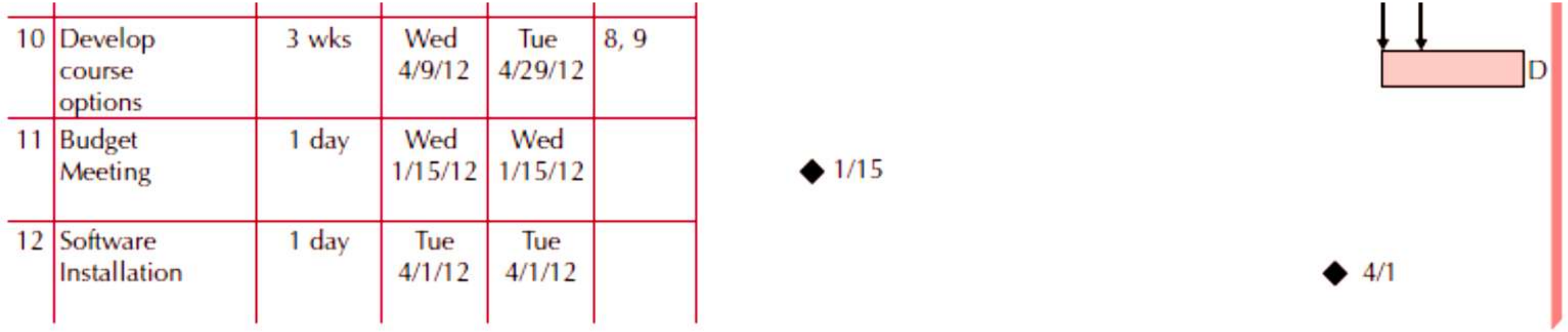
ID	Task Name	Duration	Start	Finish	Prede
1	Identify vendors	2 wks	Wed 1/1/12	Tue 1/14/12	
2	Review training materials	6 wks	Wed 1/1/12	Tue 2/11/12	
3	Compare vendors	2 wks	Wed 2/12/12	Tue 2/25/12	2
4	Negotiate with vendors	3 wks	Wed 2/26/12	Tue 3/8/12	3



GC's special notations

There are a few special notations that can be placed on a Gantt chart. **Project milestones are shown using upside-down triangles or diamonds**. **Arrows** are drawn between the task bars to show task dependencies. Sometimes, **the names of people** assigned to each task are listed next to the task bars to show what human resources have been allocated to the tasks.

10	Develop course options	3 wks	Wed 4/9/12	Tue 4/29/12	8, 9
11	Budget Meeting	1 day	Wed 1/15/12	Wed 1/15/12	
12	Software Installation	1 day	Tue 4/1/12	Tue 4/1/12	

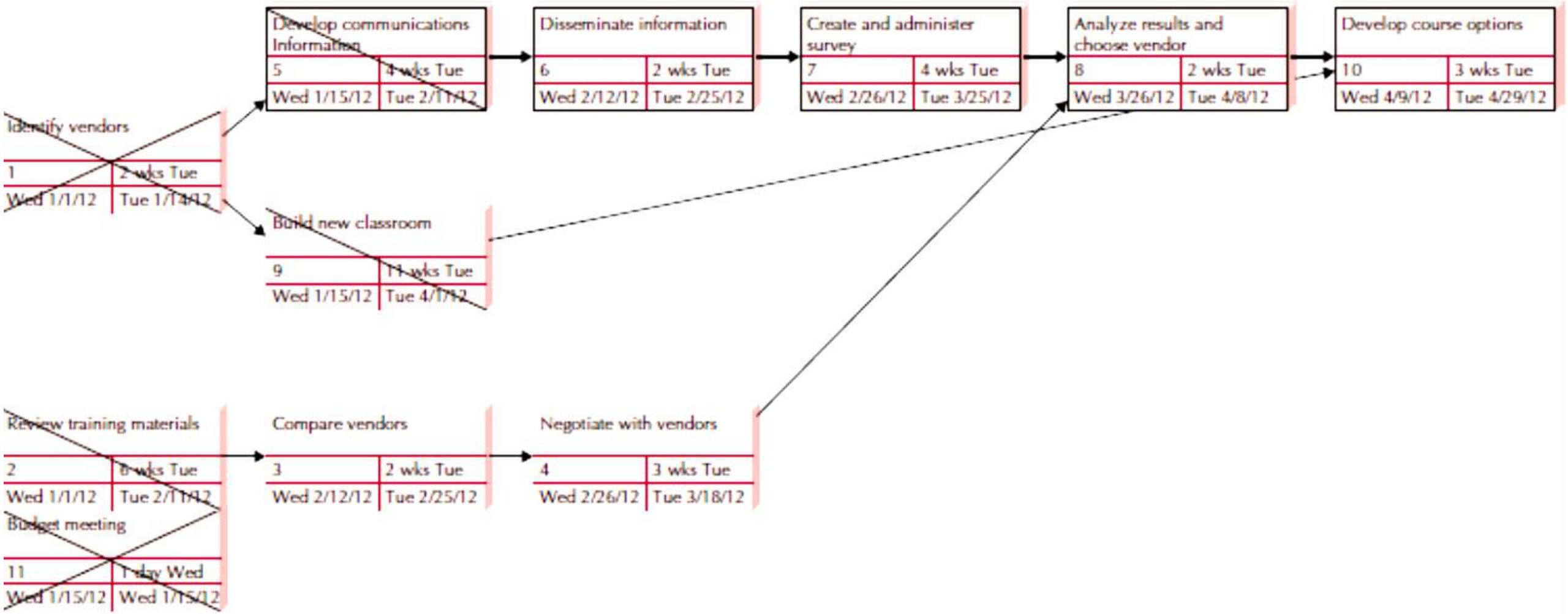


Network Diagram

A second graphical way to look at project workplan information is the network diagram that lays out the project tasks in a flowchart. Program Evaluation and Review Technique (PERT) is a network analysis technique that can be used when the individual task time estimates are fairly uncertain. Instead of simply putting a point estimate for the duration estimate, PERT uses three time estimates: optimistic, most likely, and a pessimistic.

$$\text{PERT weighted average} = \frac{\text{optimistic estimate} + (4 * \text{most likely estimate}) + \text{pessimistic estimate}}{6}$$

Network Diagram



The network diagram is drawn as **a node-and-arc type of graph** that shows **time estimates in the nodes** and **task dependencies on the arcs**. Each *node* represents an individual task, and a line connecting two nodes represents the dependency between two tasks. **Partially completed** tasks are usually displayed with a **diagonal line** through the node, and **completed tasks** contain **crossed lines**.

Critical Path Method

The *critical path method* (CPM) simply allows the identification of the critical path in the network. The critical path is the longest path from the project inception to completion. The critical path shows all the tasks that must be completed on schedule for a project as a whole to finish on schedule. If any tasks on the critical path take longer than expected, the entire project will fall behind. Each task on the critical path is a *critical task*, and they are usually depicted in a unique way; in Figure they are shown with double borders (see tasks 5, 6, 7, 8, and 10). CPM can be used with or without PERT.

Project Effort Estimation

The science (or art) of project management is in making trade-offs among three important concepts: **the functionality of the system**, **the time to complete the project** (when the project will be finished), and **the cost of the project**. Most accurate estimates come from experience.

In the beginning of the project, the manager needs to estimate each of these levers and then **continuously assess** how to roll out the project in a way that meets the organization's needs.

Estimation is the process of assigning projected values for time and effort.

Numbers

The numbers used to calculate these estimates can come from several sources. For example they can be taken from projects with similar tasks and technologies or provided by experienced developers. Generally speaking, the numbers should be conservative. A good practice is to keep track of the actual values for time and effort during the development process so that numbers can be refined along the way and the next project can benefit from real data.

CREATING AND MANAGING THE WORKPLAN

Once a project manager has a general idea of the functionality and effort for the project, he or she creates a *workplan*, which is a dynamic schedule that records and keeps track of all the tasks that need to be accomplished over the course of the project

Workplan: a dynamic and sequential list of all tasks needed to complete a project

Approaches:

- Modify existing or completed projects
- Derive the tasks from the methodology being used

Agile Process:

- Iterative & incremental

Workplan is also iterative & incremental

- Tasks and time intervals follow the phases
- Different tasks executed for each workflow

Evolutionary Work Breakdown Structures and Iterative Workplans

Because object-oriented systems approaches to systems analysis and design **support incremental and iterative development**, any project planning approach for object-oriented systems development also requires an incremental and iterative process.

In many ways, a workplan for an incremental and iterative development process is organized in a similar manner.

WBSs tend to have three underlying problems: 1st:

- They tend to be **focused on the design of the information system** being developed. The creation of the WBS **forces the premature decomposition** of the system design and the tasks associated with creating the design of the system.
- **Where the problem domain is well understood, tying the structure of the workplan to the product to be created makes sense**.
- However, in cases **where the problem domain is not well understood, the analyst must commit to the architecture of the system being developed before the requirements of the system are fully understood.**

2nd:

- They tend to force **too many levels of detail very early** on in the systems development process **for large projects** or they tend to allow **too few levels of detail for small projects**.

Because the **primary purpose of a WBS is to allow cost estimation and scheduling to take place**, in conventional approaches to planning, the WBS must be done correctly and completely at the beginning of the development process. To say the least, **this is a very difficult task to accomplish with any degree of validity**. In such cases, it is no wonder that cost and schedule estimation for many information systems development projects tend to be wildly inaccurate.

3rd:

- ***Because they are project specific, they are very difficult to compare across projects.***

This leads to ineffective learning across the organization. Without some standard approach to create WBSs, it is difficult for project managers to learn from previous projects managed by others. This tends to encourage the reinventing the wheel and allows managers to make the same mistakes that previous managers have made.

Managing Scope

An analyst may assume that a project will be safe from scheduling problems because he or she carefully estimated and planned the project up front.

However, the most common reason **for schedule and cost overruns—*scope creep*—**occurs after the project is under way.

Scope creep happens when new requirements are added to the project after the original project scope was defined and frozen. It can happen for many reasons:

- Users might suddenly understand the potential of the new system and realize new functionality that would be useful;
- Developers might discover interesting capabilities to which they become very attached;
- A senior manager might decide to let this system support a new strategy that was developed at a recent board meeting.

Timeboxing

Another approach to scope management is a technique called **timeboxing**.

Up until now, we have described task-oriented projects. In other words, we have described projects that have a schedule driven by the tasks that need to be accomplished, so the greater number of tasks and requirements, the longer the project will take.

Some companies have little patience for development projects that take a long time, and these companies take a time-oriented approach that places meeting a deadline above delivering functionality.

Steps for Timeboxing

1. Set the date for system delivery.
2. Prioritize the functionality that needs to be included in the system.
3. Build the core of the system (the functionality ranked as most important).
4. Postpone functionality that cannot be provided within the time frame.
5. Deliver the system with core functionality.
6. Repeat steps 3 through 5 to add refinements and enhancements.

Refining Estimates

The estimates that are produced during planning need to be refined as the project progresses. This does not mean that estimates were poorly done at the start of the project; rather, it is virtually impossible to develop an exact assessment of the project's schedule at the beginning of the development process.

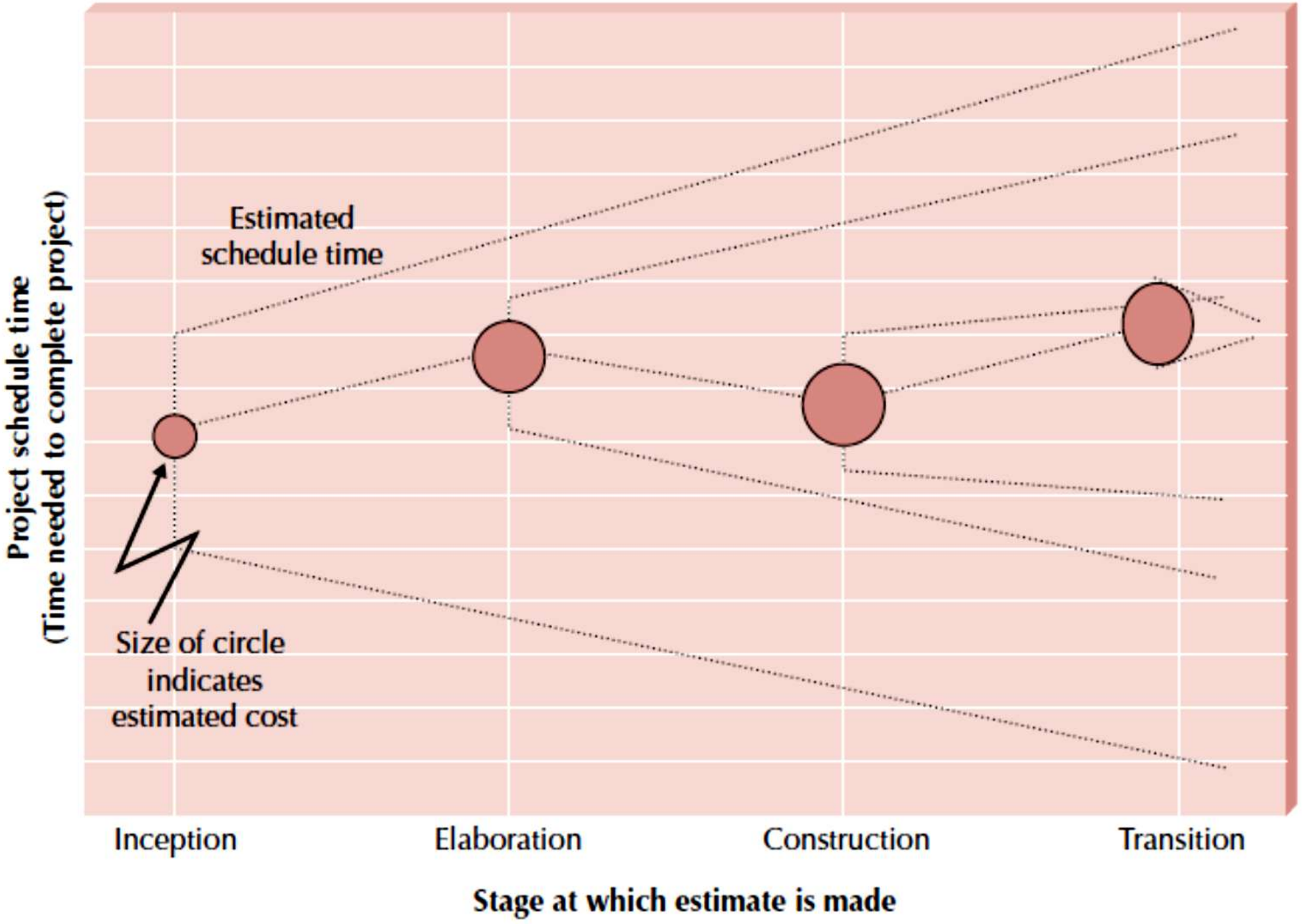
A project manager should expect to be satisfied with broad ranges of estimates that become more and more specific as the project's product becomes better defined.

Hurricane model

In many respects, estimating what an IS development project will cost, how long it will take, and what the final system will actually do follows a *hurricane model*. When storms and hurricanes first appear in the Atlantic or Pacific, forecasters watch their behavior and, on the basis of minimal information about them (but armed with lots of data on previous storms), attempt to predict when and where the storms will hit and what damage they will do when they arrive.

According to one of the leading experts in software development, a well-done project plan (prepared at the end of inception) has a 100 percent margin of error for project cost and a 25 percent margin of error for schedule time.

Hurricane Model



Schedule

What happens if you overshoot an estimate (e.g., analysis ends up lasting two weeks longer than expected)? There are a number of ways to adjust future estimates. If the project team finishes a step ahead of schedule, most project managers shift the deadlines sooner by the same amount but do not adjust the promised completion date. The challenge, however, occurs when the project team is late in meeting a scheduled date. Three possible responses to missed schedule dates are presented in Figure

Possible Actions When a Schedule Date Is Missed

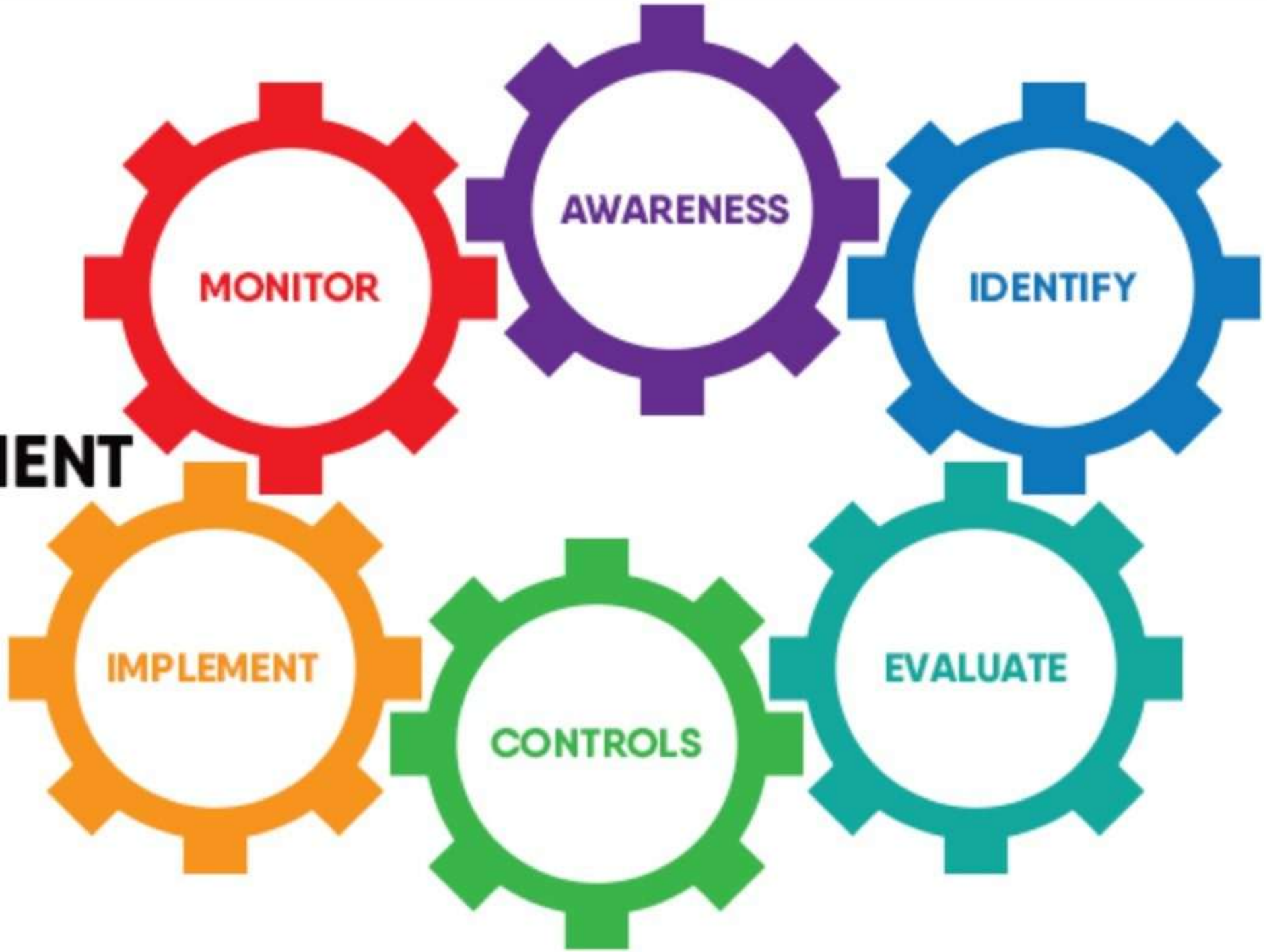
Assumptions	Actions	Level of Risk
If you assume the rest of the project is simpler than the part that was late and is also simpler than believed when the original schedule estimates were made, you can make up lost time.	Do not change schedule.	High risk
If you assume the rest of the project is simpler than the part that was late and is no more complex than the original estimate assumed, you can't make up the lost time, but you will not lose time on the rest of the project.	Increase the entire schedule by the total amount of time that you are behind (e.g., if you missed the scheduled date by two weeks, move the rest of the schedule dates to two weeks later). If you included padded time at the end of the project in the original schedule, you might not have to change the promised system delivery date; you'll just use up the padded time.	Moderate risk

If you assume that the rest of the project is as complex as the part that was late (your original estimates were too optimistic), then all the scheduled dates in the future underestimate the real time required by the same percentage as the part that was late.	Increase the entire schedule by the percentage of weeks that you are behind (e.g., if you are two weeks late on part of the project that was supposed to take eight weeks, you need to increase all remaining time estimates by 25 percent). If this moves the new delivery date beyond what is acceptable to the project sponsor, the scope of the project must be reduced.	Low risk

Managing Risk

One **final facet** of project management is ***risk management***, the process of assessing and addressing the risks that are associated with developing a project. Many things can cause risks: weak personnel, scope creep, poor design, and overly optimistic estimates. The project team must be aware of potential risks so that problems can be avoided or controlled well ahead of time.

RISK MANAGEMENT



Risk Assessment

RISK 1: The development of this system likely will be slowed considerably because project team members have not programmed in Java prior to this project.

Likelihood of risk: High probability of risk.

Potential impact on the project: This risk will probably increase the time to complete programming tasks by 50 percent.

Ways to address this risk:

It is very important that time and resources are allocated to up-front training in Java for the programmers who are used for this project. Adequate training will reduce the initial learning curve for Java when programming begins. Additionally, outside Java expertise should be brought in for at least some part of the early programming tasks. This person should be used to provide experiential knowledge to the project team so that Java-related issues (of which novice Java programmers would be unaware) are overcome.

RISK 2: …

Scope Management

Scope “creep”

- Occurs after the project is underway
- Results from adding new requirements to the project
- Can have a deleterious effect on the schedule
- Project Manager is responsible to manage changes to reduce scope creep

Techniques to manage the project scope:

- Identify all requirements at the outset
- Allow only those changes deemed absolutely necessary
- Carefully examine the impact of suggested changes
- Delay some changes for “future enhancements”
- Time boxing



JIRA - Core Features

Boards

- JIRA supports Scrum and Kanban boards.
- These boards provide an immediate snapshot of the project to the team.
- Helps to quickly review the progress of the project and see the status of the individual tasks.
- Board workflow can be customized to fulfil the way a team wants to proceed.

Business Project Template

- JIRA supports number of business templates to manage simple tasks and complex tasks like workflow.
- Template can be customized based on the team and their approach. Ex: Workflow can be customized based on each team's approach.
- Every step is accounted and team can move to achieve their goals.

JIRA - Project

An Issue is classified as follows −

Sub-Task − This is the sub-task of an issue. In a logged issue, there can be different tasks to resolve it, which are called as sub-tasks.

Bug − A problem that impairs or prevents the functions of the product.

Epic − A big user story that needs to be broken down. Created by JIRA Software - do not edit or delete.

Story − A user story. Created by JIRA Software - do not edit or delete.

Task − A task that needs to be done to achieve team's goal.